

Installing a GoTek in a SVI-738 MSX

Disclaimer: I have made this up as I found it hard to find anything that could hold your hand in doing this and was all in one place... there is a lot of information on the net and I have taken bits and pieces from all over and are not my own pictures and information. However most all are my own story on setting it up and my journey in doing so. If you need to contact me for help you can email me and I hope this helps you in your journey.. also if you feel you want to help me in doing other things like this you can give me something for the effort into my paypal account.

(same as email xpress_comp@live.com)

Ok so on with the show...

- 1- Now working with old pre and newer PC's has become a pain in the butt as most the older systems relied on Floppy Drive to store their data. Now that's fine back some 20+ years ago but now it is the drives and the media that is going bad and to get good ones to replace faulty gear is getting harder to find and the cost is getting over the top. There must be an easy way to fix this problem and until a few years back there was not much for a off the shelf fix, now step in the HxC GoTek USB drive now this is a game changer. if you want to read about the history and whom made it search on google for GoTek.
- 2- Now we are going to need some parts to do this and it's not just a put it in and off you go, there is more to it than that. We have to set it up with different software as what they come with or have on them by default is not that good, well not for what we want to use it for anyway.

Items needed; GoTek Drive: Model Number: SFR1M44-U100K,

USB serial Programmer: USB 2.0 to TTL UART Adapter Module 3.3V/5V Power.

some 20cm F-F cables 200mm:

Some header pins single row. (about a row of 10 max)



EXTRAS: Rotary encoder,

OLED module 0.91" white or blue OLED 128X32 OLED LCD LED Display module.



Also needed is a solder iron, solder, side cutters and a Philips (cross head) screw driver.

- 3- Now the standard GoTek comes with a 3 x 7 segment display, 2 buttons for up and down selection, activity led and the USB port on the front.



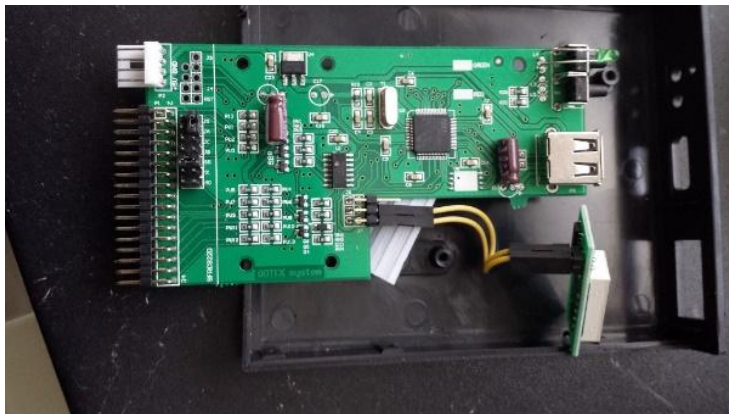
- 4- On the back is where the 34 pin Floppy header, standard 3.5" power connector and the drive setting jumpers. You will also see a set of header holes where some pins were not installed behind the power connector.



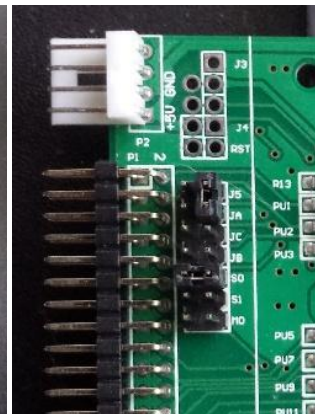
- 5- Ok so what you need to decide is, do you want to just live with the standard display or upgrade it to the Oled display? The extra cost is not much at all around \$4.00 plus p&p for the Oled and encoder. You can buy most items from **Aliexpress.com** or **Wish.com** and the difference in information you see I think it is worth doing. If you don't want to do the upgrade then just skip over that part.
- 6- Now let's start, open up the unit by removing the 3 screws on top and put them some where safe. The board and display are just clipped in so you can press on display and pop it inside the case also press the front led inside the case and then gently lift the back of the board up and pull back (the USB goes through the plastic front) and the board should come out easy.



- 7- So, we have the unit out you can see the block of jumpers behind the 34-pin floppy connector and the 2 x rows that were not put in. the ones that were not put in are needed if you want to program the drives firmware with the USB serial programmer. (you can do it with a USB A to A cable from the front) but I won't cover this here sorry.

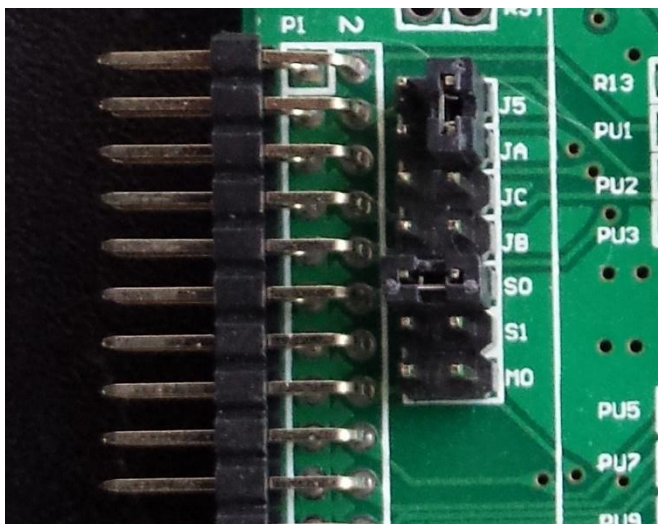


(the GoTek board out of case)

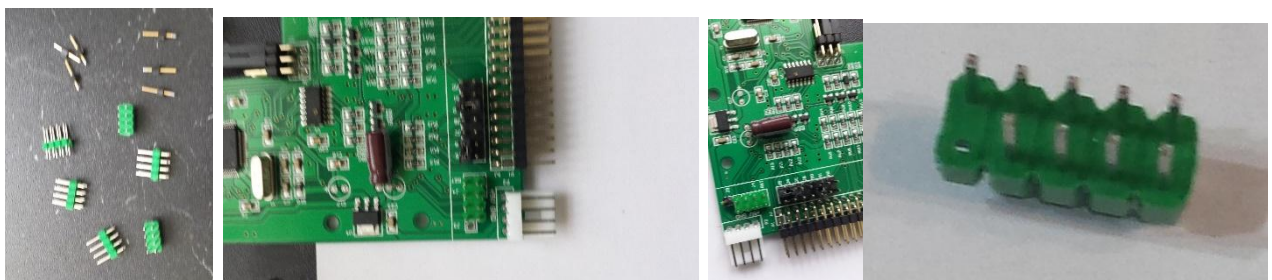


(jumper blocks)

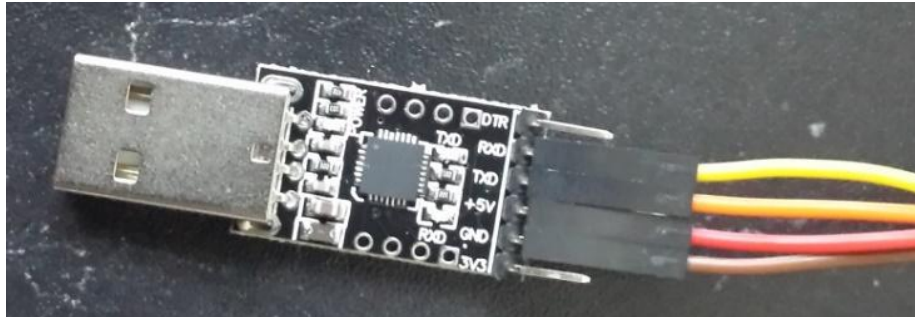
- 8- Also, some drives have the power led on the board and just left inside and some didn't have it but we won't worry about it as the display will tell us when power is on in the drive. Well the jumpers in the block behind the floppy cable are used for drive setting's and these are listed below, but the only one you need to worry about is S0 and S1 that are the drive selectors for drive A and B. select the drive you need for the 738, internal it is S0 for an external S1. don't worry about the jumper across j4 and j5 we need to remove it before we program the unit and put it elsewhere. (it is a spare jumper)



- 9- Now the block that has no pins in them, we need to solder some so we can program the unit. Take a strip of 40-way single header strip and cut/snap off a set of 5 pins and a strip of 3 or 4 pins depends if your unit has 3 or 4 holes. Put them in and solder in place. I had some 2 row strips I used with some singles I used at the end (the back one). Another way is to cut a strip of 2 x 5 and take the 1 pin out that's not used.

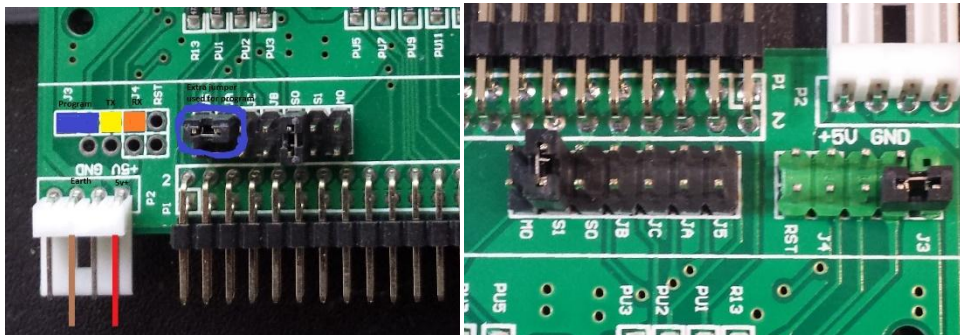


- 10- With this done what we need to do is use the jumper wires to set it up to be programmed. When we do this, you have to remember that the TX and RX only are crossed over on the board to the programmer. That meaning the TX from the board goes to the RX on programmer and the RX on the board goes to the TX on the programmer. As TX=send and RX=receive.



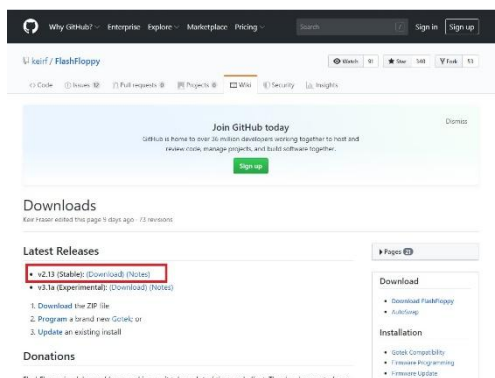
(USB serial programmer with cable on, yellow and orange crossed over.)

- 11- The next thing to do is to put a wire from the 2nd pin on the left and the last pin on the right. The 2nd on the left is earth and the right is 5v+ for programming, the 2nd wire on left goes to earth on programmer and the right wire to 5v+ on programmer. Now we need a jumper to cover 2 pins that are inline on the left top next to the TX/RX wires. (blue block on left) my GoTek had a spear jumper on the board. (circled in blue)

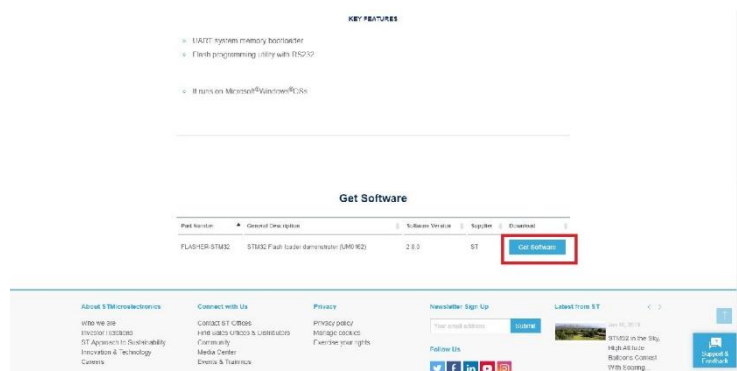


(program jumper on)
(my color system is brown=earth, red=5v, yellow=TX, orange= RX, Blue=extra Jumper for program)

- 12- Now we are ready to program the unit. We are going to use a software called flash-floppy it is better and more friendly than the one on the drive when you bought it and it supports the OLED display.
So we need to download some software, so go to google and type in :
<https://github.com/keirf/FlashFloppy/wiki/Downloads> this will take you to the makers site and is updated from time to time. Select the newest full release not the beta and download it. Once you have that Download the USB Programmer software from <https://www.st.com/en/development-tools/flasher-stm32.html>, scroll down the bottom and get the download file at the bottom of the page.

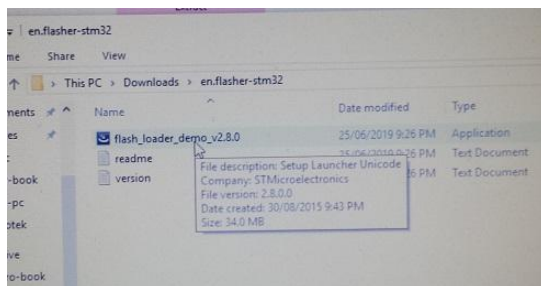


(flashfloppy stable download)

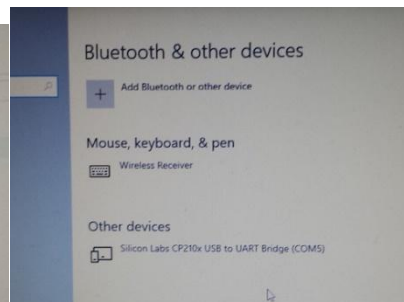


(stm32 programer download)

- 13- Ok so now you should have the GoTek connected to the USB programmer and have down loaded the 2 pieces of software we are going to need to set the unit up. Unzip both software in to folders on your desktop for easy access or where every you want. Now plug in the USB programmer and you should hear the drive connect through windows and windows will set up the USB port to be used. Check in control panel that it has.

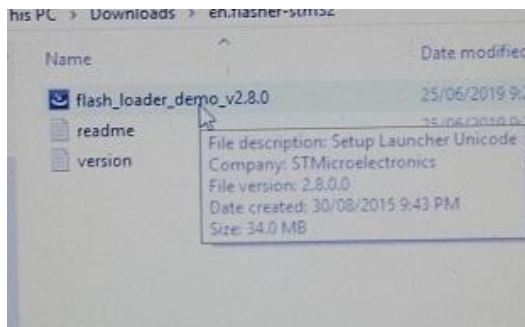


(folder with software)

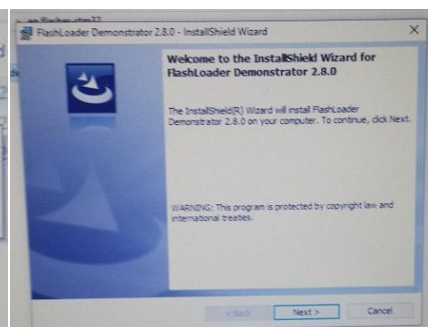


(is port listed?)

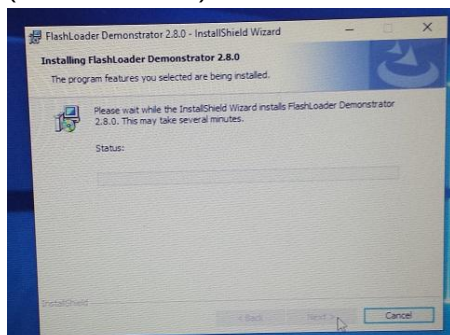
- 14- With the software unzipped go to the programmer folder and run the STM32 program (you may need to run as administrator), check that the programmer is set to the right USB port in the first drop-down box up top. You don't want to try program to the wrong drive now do we. If you don't see any com ports you will need to down load the drivers for the version of windows you have, either 32 or 64bit and install them.



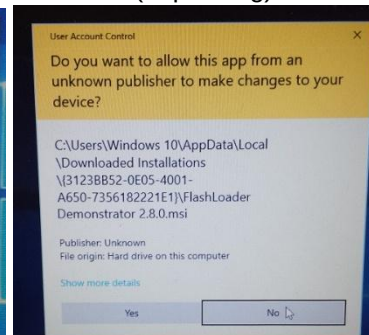
(install software)



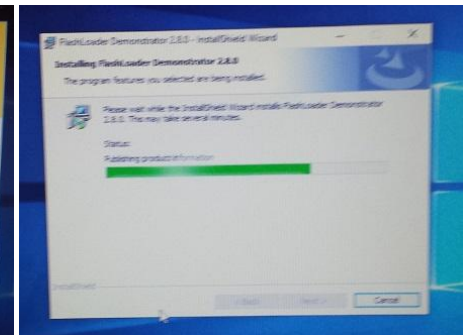
(unpacking)



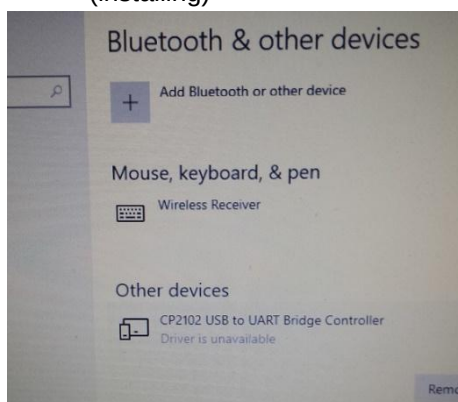
(installing)



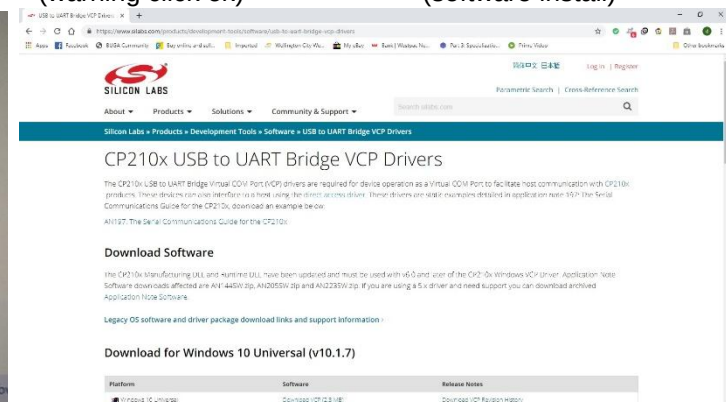
(warning click ok)



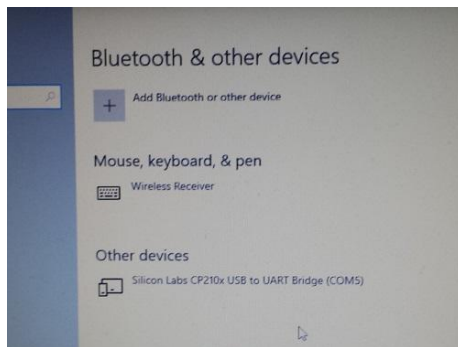
(software install)



(driver needed)



(Driver down load)

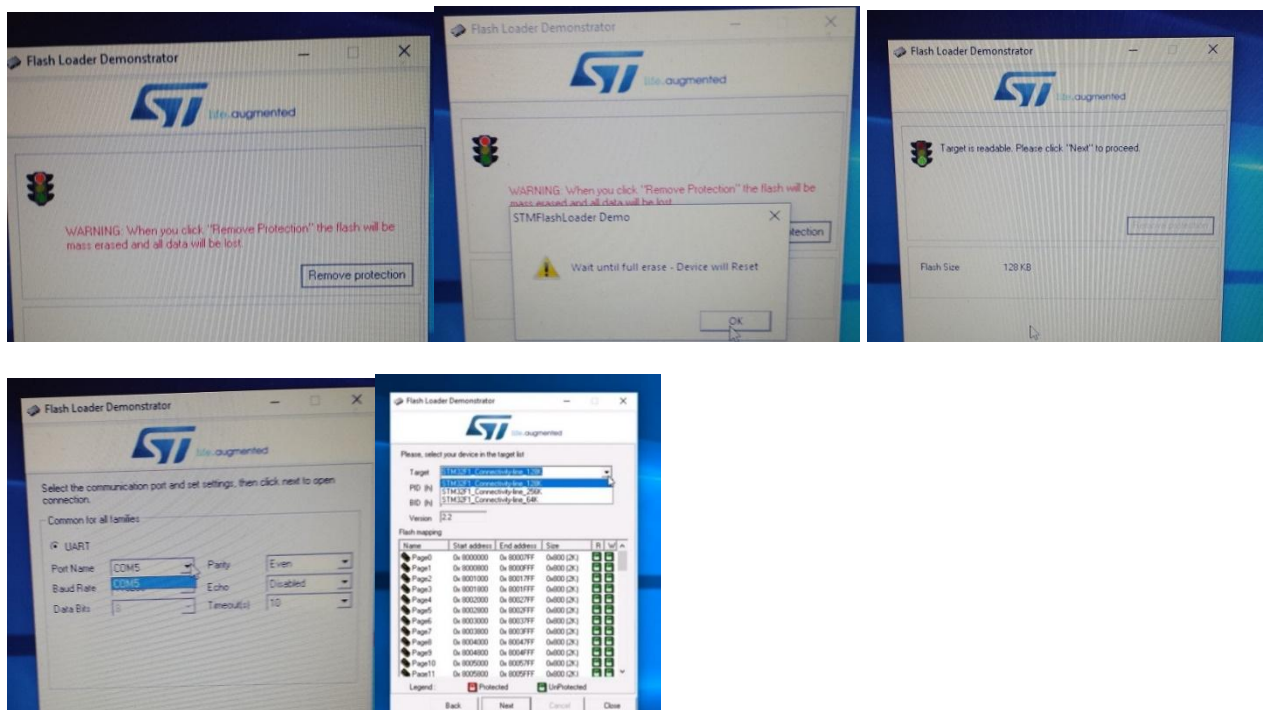


(Driver now installed)

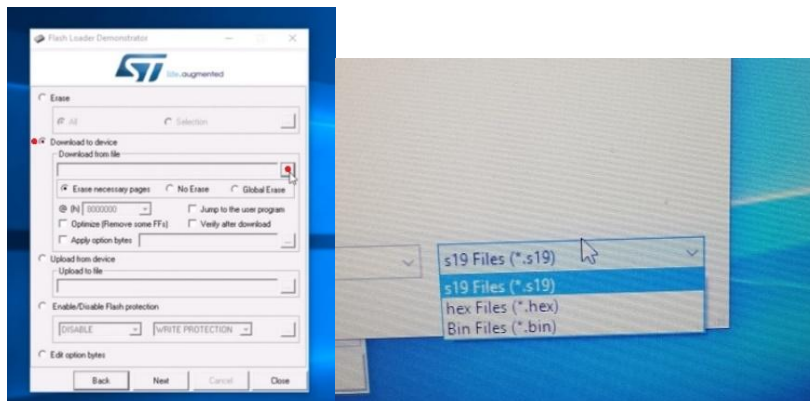


(port listed now)

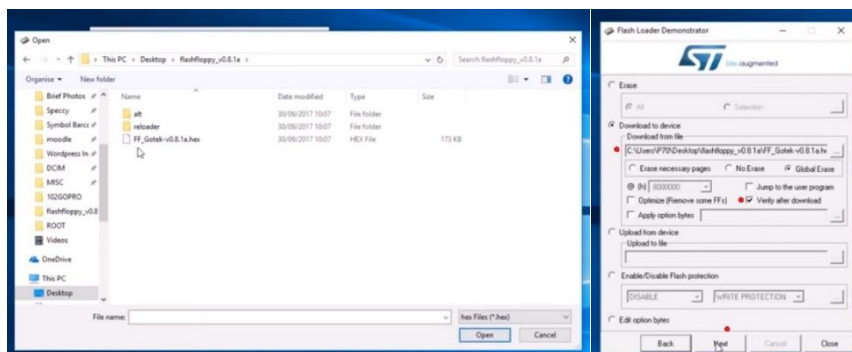
- 15- Now with software started if it is the first time it has had flash floppy installed you will need to select unprotect drive. You only need to do this the first time, after this you can update from a USB stick at any time with Flash floppy upgrades without having to remove the drive. Now click next and a config screen should come up. Click the top box to make sure the right USB is selected.



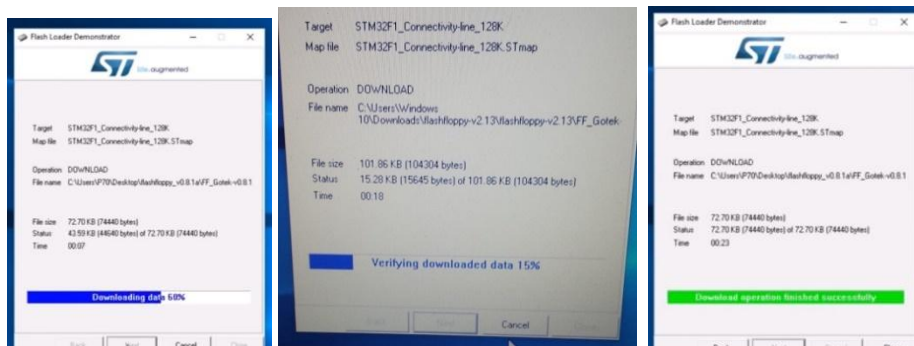
- 16- With this all done now click on next and the next screen you should see is where we select Download to Device and then select the download file box and this is where you need to go to the folder where the flash floppy files are. When you get to the folder you may not see the file and need to select show *.hex files on bottom box.



- 17- When you see the flash floppy file double click on it to select and insert it into the programmer software. You should see it in the files box now, select the verify option as well. Now select the next button.

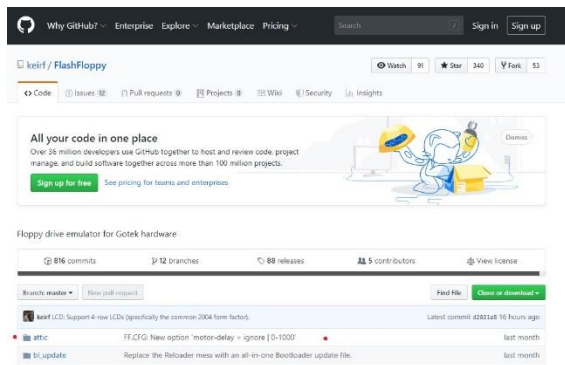


- 18- When the next button is selected the programmer will delete and upgrade the drive unit, then it will verify the software. When it finishes the bar will turn green.

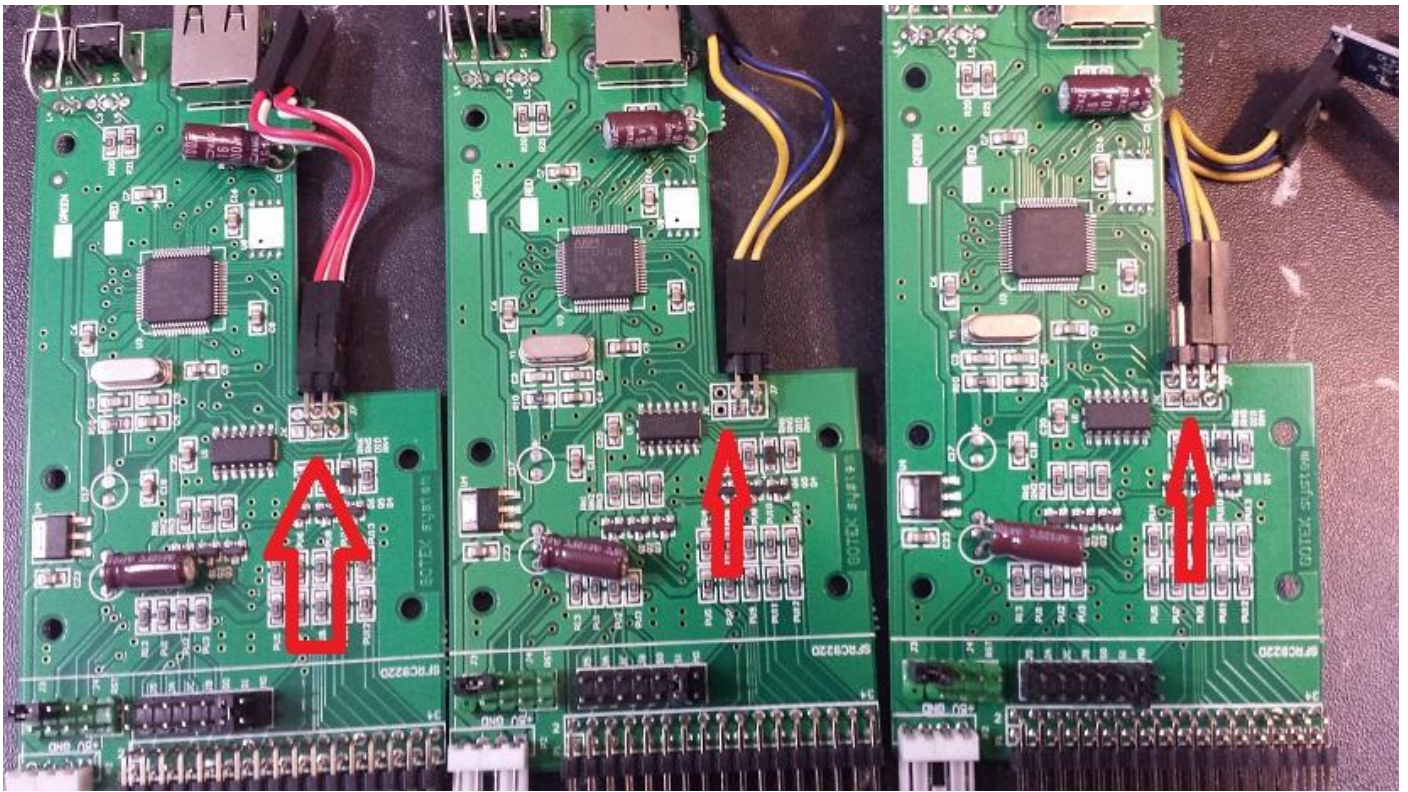


- 19- With this done and the bar Green showing go and click close. We have now finished programming the drive with the firm ware what we need to do now is setup a USB pen drive (USB stick). So, get yourself a USB pen drive up to 32Gb max but most you would need is 2-8Gb and these are cheap on wish or AliExpress (just a few dollars). Format the Pen drive in fat-16 or fat-32 only (you can download 32format off the net)

- 20- With the drive formatted to Fat-16 or fat-32 you now need to download a config file from this sight; <https://github.com/keirf/FlashFloppy> and download the file **FF.CFG** as marked in picture with red dots. That is if it is not in the Flash Floppy folder already in the examples folder when you un-zipped it.



- 21- Now we need to alter this file to run on the MSX system. A list of all the switches in the file is at the end of this or on the official web site. Also, a big thankyou to **Keir Fraser** for the firmware (Flash Floppy)
- 22- What we need to do is open the FF.CFG file in a text editor like Notepad (other word editors may alter the file) with it open look for the line and replace **"host = unspecified"** with **"host = msx"** (in lowercase), otherwise the DSK 360kb files will not work as this gives 1DD support and save as txt file type.
- 23- What we now need to do is save the file on to the USB pen drive and when done un-plug and put it to one side then un-plug the USB programmer from the pc. Unplug the cables off the Drive and take the jumper off the end block as it is only for programming the drive,
- 24- For now, sit the GoTek in its bottom case and plug the flash drive (pen drive) in the front. Now plug in the floppy cable and watch the way around the cable goes (pin-1) and the power cable. Now turn on the 738 and let the system boot up. You should see the led come up F-F then 000. (as we don't have any Disk images on a pen drive on the drive)
- 25- We don't have any software on the GoTek as of yet but will come to that after the next part so next we are going to install the Oled and rotary encoder.to the GoTek, for now remove the USB pen drive from GoTek as we don't need the FF.CFG file now as it has installed on to the drive internals.
- 26- **NOTICE** not all GoTek are made equal... some have different color wires to the led display while others don't have the pins installed for the rotary encoder. All were ordered at the same time so just watch out and if needed solder in the header pin if putting in encoder. If not don't worry about it.



(has different colors and not pins)

(normal colors and no pins)

(normal and has pin)

Oled and Rotary Encoder

Items needed:

The Oled display listed at the beginning

The Rotary Encoder and a knob.

4 x jumper leads (200mm) and some spar wire.

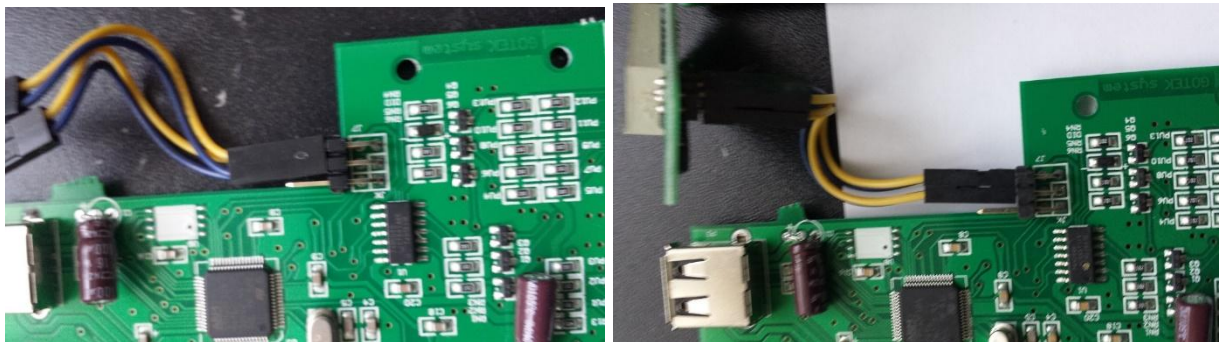
Extras; a drill and bits, box cutter (Stanley knife), hot glue gun and a soldering iron & solder.

I've seen the light!

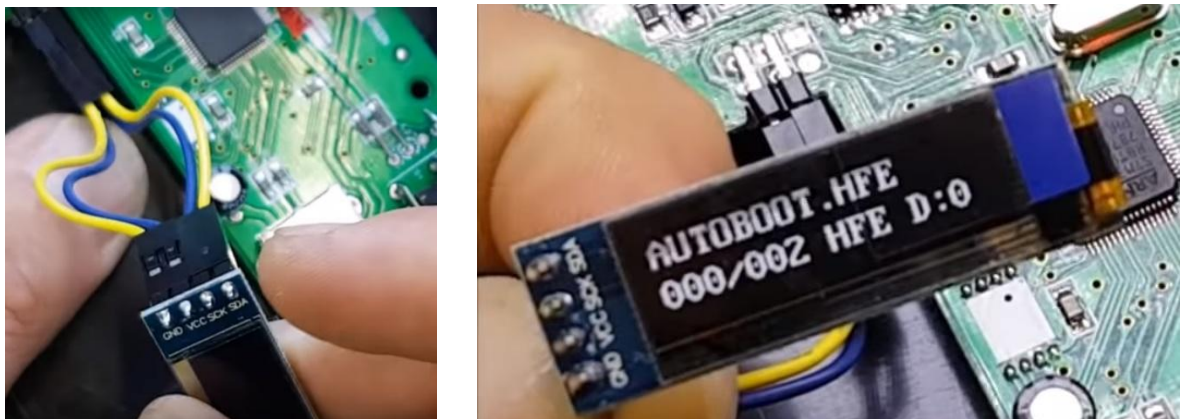
- 1- Ok so what you may have cotton is one of 2 Oled displays and maybe the rotary encoder was on a board. If this is so it's not the end, you may have to solder the strip in the one not attached on the display. And the encoder if on a little board can be de-soldered from it as the board is not used.



- 2- Now with that sorted pick up the drive board and look at the cable going to the front display, I hope you haven't taken it off yet..... only joking. What we want to do is remove the led display but leave the cables on the drive (mine are 2 x dual Yellow and blue wire). What we now want to do is remove the cables from the header and rotate them right as in the pictures, they go from across the pins to up and down the pins.

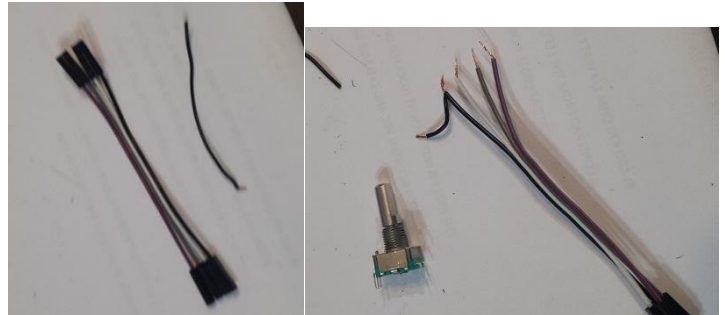
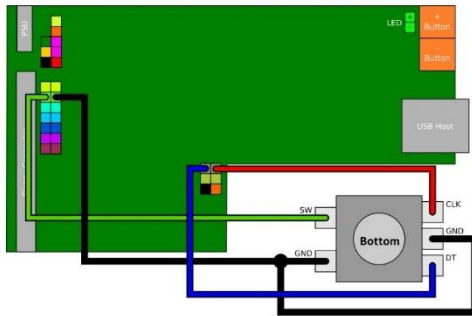


- 3- With that done now pickup your Oled display and plug it in like this, the now right-side yellow goes to SDA, the blue under it goes to SCK, the yellow on the right goes to GND and the blue under it goes to VCC. That's it done.



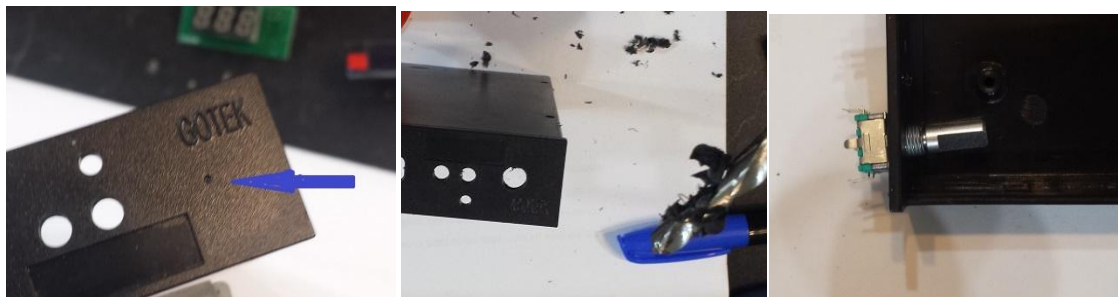
- 4- Ok so now for the encoder, so take the encoder and look at it from the bottom with the pins pointing towards you. (Pin-1=red, pin2=black, pin-3=blue, pin-4=black, pin-5 green) so pin1 goes to the bottom, pin-3 goes to the top. Pin-3 goes to the left of JA pin-2 goes to pin-4 then goes to the pin JA. Now solder in wires to the color you want to use. Don't get them mixed up. I use 4 x jumpers like I used in the programmer and just cut a set of plugs on one end only. You need the plugs on the other end to plug in to

the board. And just use a piece of wire to go from pin-2 to pin-4. Your wires maybe different colors that you use so be careful and sub the picture colors for yours.

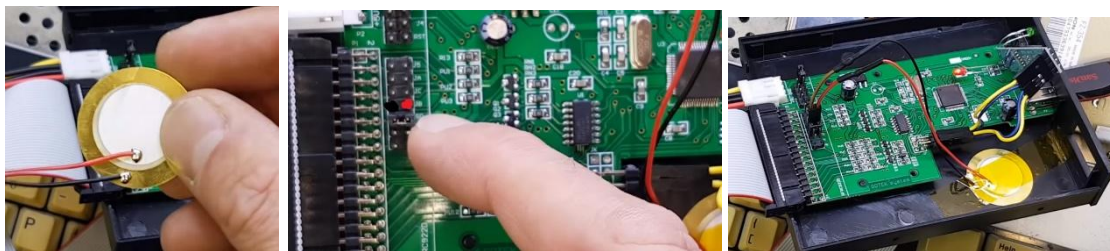


(the colors I used are; 1-purple, 2-black, 3-grey, 4-black, 5-white. Make sure you have yours set right)

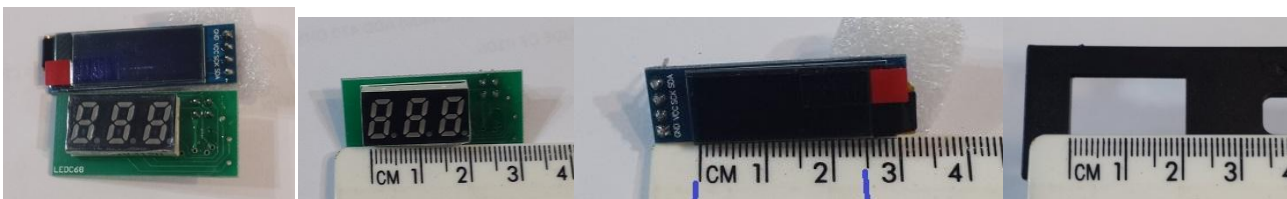
- 5- To attach the encoder, look at the GoTek and you will see on the right a small indent where a hole was meant to be. With a 6mm drill make a hole. Be slow so as not to damage the case when drilling. If it is too small use a 7mm drill to enlarge it don't install the encoder yet just see if it fits.



- 6- Now if you want the old school drive click,click,click as it reads there is a simple mod to do this also. You need a micro speaker or piezo speaker. Solder 2 wires to the speaker with 2 pin sockets at the end. With that what we do is plug in the speaker into the pins at the back behind the floppy connector. The pins they go on are the ones just above the S0 (drive select jumper) it doesn't matter what way round it goes. You can tape the speaker to the top or bottom. See the middle picture for the pins used for the speaker.



- 7- Now to finish off you may need to alter the hole that the original display was in as the Oled is bigger, if so, use a Stanley knife.to make it bigger so the screen just fits. (about an 1in wide it needs to be done). Put the screen in and see if it looks ok, if it looks ok then use very little hot glue, blue-tac, etc. to hold it in place. the pins go to the right looking at the front of the drive and point to the back of the case.

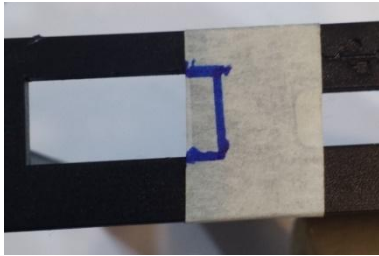


(the 2 displays)

(led display 2cm)

(Oled display 2.5cm)

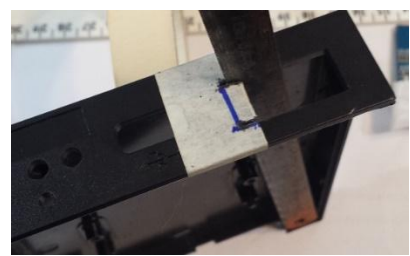
(display case hole 2cm)



(masking tape to mark area to cut out)



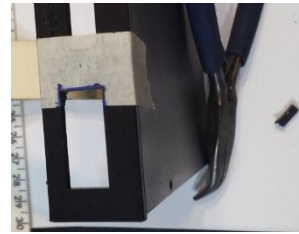
(hacksaw to cut sides)



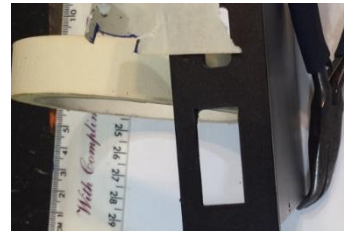
(stop just at the lines)



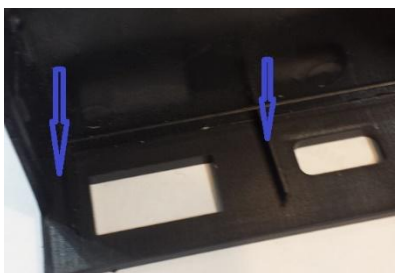
(score across to meet 2 cuts)



(with plyers gently beak out)



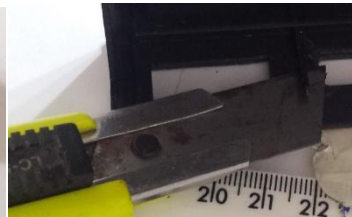
(you may have to clean a little)



(old braces for lcd)



(put tape over, set Oled and mark Depth from window top)



(mark both braces and with knife cut down to the front, remove tape and gently cut down to cuts done before. Now break out the old braces and clean up. You may need to do a little to get it looking good. check from front)



(Oled sitting in good now)



(remove the plastic off Oled – pull red tab and hot glue Oled in place)

- 8- With the Oled in place put the Encoder in case and do up the nut in front and press the knob onto the encoder. You may notice the encoder pushes in that is normal as it is the select/eject button for the disk images we will use. With that done sit the main GoTek board on the case but not seated down as we need to put the cables back on and makes it easier.
- 9- Now is the time to double check every thing is connected right. If so, gently put the led in its hole in the front and put the USB and buttons in their holes and clip it down. Check just to be sure nothing has come off and if not put the lid on and screw in down. And tools down we are done.
- 10- The last thig to do is to connect up the GoTek drive and boot the system up on the 738. If all goes well you have a working GoTek on an SV-738. You can make your own mount for inside the case. The only thing you need to do is put some DSK MSX files on the USB pen drive and have fun.

Running software and writing to the GoTek

- 1- To use the GoTek on the sv-738 MSX you will need a USB pen drive of any size up to a max size of 32Gb at the time of this tutorial. You will also need to format it to fat-16 or fat-32 on a PC or mac etc..
- 2- With that USB drive formatted you can copy over the DSK files you want to use, just put them in the root directory of the Pen drive while it is plugged into the pc.
- 3- Now plug in the USB to the GoTek and turn on the sv-738 system now you should see the first DSK file listed and if you turn the encoder knob left and right the files will change. When you have the disk file showing that you want press the knob and that loads the file. If you press in again it Ejects the file.
- 4- If you are using the un-mod version of the GoTek then you will use the 2 buttons for up and down scrolling of the files and pressing your reset.

FF.CFG Configuration File

Flash Floppy has many configurable settings which can be specified in an INI-style configuration file called FF.CFG, placed in the root folder or FF/ subfolder of your USB drive. An example is provided in the examples/ subfolder of the Flash Floppy distribution.

Defaults

Out of the box, Flash Floppy loads the default values described in the Options list, below. When these are modified by values in FF.CFG they are recorded into the Gotek's Flash memory and become the new default values when the drive is next powered on.

If you wish to return to 'factory defaults', press the Prev and Next buttons (or Select, or the rotary encoder, if you have them) for three seconds with no USB stick inserted. The display will show "RST" or "Reset Flash Configuration", and Flash Floppy will return to factory defaults when the buttons are released.

Options

Default values are marked by asterisk.

Drive Emulation:

interface = shugart | ibmpc | ibmpc-hdout | akai-s950 | amiga | jc*

Pin assignments of the floppy-drive interface

shugart: Shugart interface (Amiga, Atari ST, many others)

ibmpc: IBM PC interface, no connection on pin 2

ibmpc-hdout: IBM PC, high-density-select output (Gotek->host) on pin 2

Do not select this option unless explicitly required for your system!

It is not required for IBM PC compatibles. Use ibmpc instead.

akai-s950: Akai S950

amiga: Drive ID hack on pin 34. Use shugart instead when possible.

See Amiga-specific hints for advice on this setting.

jc: Specified by jumper JC (closed = IBM PC, open = Shugart)

host = unspecified* | acorn | akai | ...

Host platform: Improves image-format detection for generic types such as IMG and DSK

acorn: Acorn ADFS

akai: Akai synths (S01, S20, S950), Korg synths, SC Prophet 3000

casio: Casio synths (FZ-1)

dec: DEC (RX33, RX50)

ensoniq: Ensoniq synths (ASR/TS series, and others)

fluke: Fluke (9100)

gem: General Music (S2, S3, S2R)

kaypro: Kaypro

memotech: Memotech

msx: MSX

nascom: Nascom (1, 2)

pc98: NEC PC-98

pc-dos: PC DOS Format (geometry determined from Bios Parameter Block)

tandy-coco: Tandy Color Computer (CoCo)

ti99: TI-99/4A

uknc: UKNC, DVK (Soviet PDP-11)

unspecified: Detection based on image-name suffix only

pin02 = auto* | nc | low | high | rdy | nrdy | dens | ndens | chg | nchg

Manually assign an output signal (Gotek->host) to floppy interface pin 2

auto: Automatically determined from interface = setting

nc: Unused / No Connection

low, high: Constant low (0v) or high (5v) voltage

rdy, nrdy: Drive ready, or logical complement

dens, ndens: Density mode (HD = 0v), or logical complement

chg, nchg: Disk changed, or logical complement

pin34 = auto* | nc | low | high | rdy | nrdy | dens | ndens | chg | nchg

Manually assign an output signal (Gotek->host) to floppy interface pin 34

write-protect = yes | no*

Are images write protected when initially mounted?

Protection can be toggled by holding eject for 2 seconds

yes: Forcibly write-protect images

no: Respect the FAT read-only attribute

side-select-glitch-filter = 0-255 (0*)

Filter glitches in the SIDE-select signal shorter than N microseconds

Useful on some old hardware (eg. CP/M systems)

track-change = instant* | realtime

Rotational offset of data after a track change

instant: No rotation during track change

realtime: Emulate rotation of disk while track is changing

index-suppression = yes* | no

Are index pulses suppressed when RDATA and WDATA inactive?

Older systems may depend on constant index pulses (eg. BBC Micro)

head-settle-ms = 0-255 (12*)

Milliseconds from head-step start to RDATA active

motor-delay = 0-1000 | ignore*

Milliseconds delay from motor-on to ready (or, ignore motor signal)

Note that on a standard unmodified Gotek the motor signal is not physically connected, and modifying this option will have no effect. See Hardware Mods to rectify this.

Further note that most systems and software do not care about correct motor emulation, and this option can be safely left at its default.

If correct emulation is required, 200 is a sensible starting value.

chgrst = step* | pa14 [v3.x]

Select action which causes the Disk Changed (DSKCHG) signal to reset

step: Step command received

pa14: Disk Change Reset signal (eg. pin 1 on some old Sony drives)

Signal is connected to SWCLK (PA14) header pin

See Hardware Mods for more details

Startup & Initialisation:

ejected-on-startup = yes | no*

Disk image loaded or ejected at power on

image-on-startup = last* | static | init

Which image (or folder) is selected at startup

last: Last-selected item at power off (recorded in IMAGE_A.CFG)

static: Static pathname specified in IMAGE_A.CFG

init: First item in root folder

display-probe-ms = 0-65535 (2000*)

Time in milliseconds to attempt to probe an attached display

If you have a 2-digit LED display, or no display: Set to 0 for faster startup

Image Navigation:

autoselect-file-secs = 0-255 (2*)

Auto-select the current file/slot after N seconds

N=0: disable auto-select

autoselect-folder-secs = 0-255 (2*)

Auto-select the current folder after N seconds

N=0: disable auto-select

folder-sort = always* | never | small

Choose how folder entries are sorted in native navigation mode

always: Always sort folder entries: Large folders may be truncated

never: Never sort folder entries, instead presenting them in FAT order

small: Only sort folders which are small enough to sort in full

sort-priority = folders* | files | none

Priority of files vs subfolders when sorting folder entries

folders: Folders are listed before files

files: Files are listed before folders

none: Folders and files are not differentiated

nav-mode = native | indexed | default*

Navigation mode for selecting images or slots

native: Navigate through all valid images and folders

indexed: Navigate through DSKA0000, DSKA0001, ...

default: As native unless overridden by HxC-compatible config

nav-loop = yes* | no

When navigating slots or a folder, loop at start/end of the list

twobutton-action = zero* | eject | rotary | rotary-fast | reverse

Actions of first two buttons

Multiple values can be specified, separated by commas (eg. rotary,reverse)

zero: 1: Prev, 2: Next, Both: Slot 0

eject: 1: Prev, 2: Next, Both: Eject/Insert

rotary: 1: Up-dir, 2: Select/Eject/Insert, Both: -

rotary-fast: 1: Prev (accelerated), 2: Next (accelerated), Both: Up-dir

reverse: Reverse operation of the buttons (useful if installed upside down)

rotary = none | quarter | half | full* | trackball | buttons | reverse

Type of input sensor at the rotary-encoder inputs (pins PC10 and PC11)

Multiple values can be specified, separated by commas (eg. half,reverse)

none: No input sensor is connected

quarter, half, full: Rotary encoder, identified by fraction of a Gray-code cycle performed per detent/click

Select half if default value (full) requires 2 clicks per move

Select quarter if default value (full) requires 4 clicks per move

trackball: Blackberry-style trackball

buttons: Push-to-ground Prev/Next buttons

reverse: Reverse direction of the knob (useful if wired backwards!)

indexed-prefix = quoted-string ("DSKA"*)

Prefix for image names in indexed navigation mode

Empty string ("") means no prefix

Display:

display-type = auto* | lcd-NNx02 | oled[-128xNN][-rotate][-narrow[er]][-ztech]

auto: Auto-detect (7-seg LED, LCD, OLED)

lcd-NNx02: NNx2 backlit LCD with I2C backpack (16 <= NN <= 40)

oled-128xNN: 128xNN I2C OLED (NN = 32 | 64)

-rotate: OLED view is rotated 180 degrees

-narrow[er]: OLED view is restricted to Gotek display cutout

-narrow: 18 characters of 6x13 font

-narrower: 16 characters of 6x13 font

-ztech: ZHONGJY_TECH 2.23" 128x32 SSD1305 OLED display [v3.x]

Must be explicitly configured as cannot be automatically detected

oled-font = 6x13* | 8x16

Select 6px- or 8px-wide OLED display font

6x13 font permits:

More characters per row

Use of G0tek display cutout (eg. display-type=oled-128x32-narrow)

oled-contrast = 0-255 (143*)

OLED contrast/brightness

oled-text = default* | list-of-character-pairs [v3.x]

Text row heights and arrangements on OLED display

Comma-separated list, one entry per OLED row, top to bottom

Each list item is a character pair: [content-row][height]

content-row: 0-2 selects specified content row, 7 selects a blank row

height: s = single height (16px), d = double height (32px)

default: Selects a sensible default dependent on display size

128x32: 0s,1s

128x64: 0d,1s,2s

display-off-secs = 0-255 (60*)

Turn LCD/OLED display off after N seconds of inactivity

N=0: always off

N=255: always on

display-on-activity = yes* | no

Automatically switch LCD/OLED display on when there is drive activity

display-scroll-rate = 100-65535 (200*)

LCD/OLED long filename scroll rate (ms per update)

Larger value means slower scroll

display-scroll-pause = 0-65535 (2000*)

LCD/OLED long filename scroll start/end pause (ms)

N=0: Endless looping scroll

nav-scroll-rate = 0-65535 (80*)

LCD/OLED long filename scroll rate during navigation (ms per update)

nav-scroll-pause = 0-65535 (300*)

LCD/OLED long filename pause before scroll starts during navigation (ms)

Miscellaneous:

step-volume = 0-20 (10*)

Speaker volume (connected at jumper JB) when drive heads are moved

da-report-version = quoted-string

Report the specified version number to host software

Empty string ("") means report the firmware version number

For example: da-report-version = "v3.0.0.0"

extend-image = yes* | no

Automatically extend a truncated image file at mount/insert time

Applies to SSD, DSD, TRD images only

<https://github.com/keirf/FlashFloppy/wiki/Downloads>

<https://www.youtube.com/watch?v=-K31S2xqZIk>

SDCard support

- SD Card up to 2GB.
- SDHC Card supported up to 32GB.

SDCard Filesystem

- FAT12/FAT16 & FAT32 supported. Subdirectory and long name file supported.
- Multi-Volumes/Images support:
 - > More than 8000 HD (1.44MB) Floppy Images on a 32GB SDCard !
 - > More than 16000 DD (720KB) Floppy Images on a 32GB SDCard !

Read / Write support

- Track mode based floppy emulator. (Full track pre-encoded in the HFE image file)
- Read support : Most of existing formats supported. (FM/MFM/GCR/Amiga/E-mu track...)

- Custom tracks supported.

- Write support : ISO MFM/DD 128/256/512/1024/2048/4096/8192Bytes sector write supported.

- Write support : ISO FM /SD 128/256/512/1024/2048/4096Bytes sector write supported.

- Any data mark supported : 0xFB, 0xF8, 0xF9, 0xFA

- Amiga Write support.

- E-mu (Emulator I / II / SP1200) Write support.

- Low level format support :

The host machine can format the image with the right disk setup exactly as with a real floppy disk ! No need to already have an image with the right format, just make it by formatting it !

Floppy bitrate supported

- 125/150/250Kbits/s (FM/SD floppies)

- 250/300Kbits/s (MFM/DD floppies)

- 500Kbits/s (HD floppies)

- (others special bitrates support available)

(Note : Variable bitrate not supported by this hardware. So protected floppy disk image (IPF and STX file format) support is only partial ! If you are looking for a device supporting IPF / STX please have a look to the USB HxC Floppy Emulator device)

RPM

- 300 RPM, 360 RPM supported.

- (others special RPM available/supported).

Tracks / Side

- Up to 255 tracks per floppy.

- 1 or 2 sides.